The kernel method on odd toroidal grids

Siddharth Narayan Mishra
working draft of a section for the joint $C_3\Box C_n$ paper

October 26, 2026

Abstract

For odd m, n the Alon–Tarsi number of $C_m\Box C_n$ is 4, so the polynomial method cannot prove 3-choosability. We examine the other standard tool, the kernel method of Bondy–Boppana–Siegel and Galvin. For $m = 3$ we settle it completely: for odd n, $C_3\Box C_n$ has a kernel-perfect orientation with maximum out-degree 2 if and only if $3 \mid n$, and then there are exactly 12 of them. The "only if" direction has a hand proof: every other orientation contains an explicit induced odd directed cycle. The "if" direction is a finite computer-assisted proof valid for all n. For $m = 5$ the method fails already on $C_5\Box C_5$. So the kernel method recovers 3-choosability of $C_3\Box C_n$ exactly when $3 \mid n$, which explains why a different method is needed.

1 Setting

$G = C_3\Box C_n$, n odd. Vertex (r, c) has row $r \in \mathbb{Z}_3$ and column $c \in \mathbb{Z}_n$. Each column is a triangle, each row an n-cycle. A *kernel* of a digraph is an independent set S such that every vertex outside S has an arc into S; a digraph is *kernel-perfect* if every induced subdigraph has a kernel.

Lemma 1 (Bondy–Boppana–Siegel)**.** *If D is a kernel-perfect orientation of G and $|L(v)| > d_D^+(v)$ for all v, then G is L-colourable. In particular a kernel-perfect orientation with $\Delta^+ \le 2$ shows that G is 3-choosable.*

G has $3n$ vertices and $6n$ edges, so $\Delta^+ \le 2$ forces $d^+(v) = d^-(v) = 2$ for every v. We call such orientations *admissible*.

Remark (Richardson never applies). Each admissible orientation is Eulerian on a connected graph, hence strongly connected. If it had no odd directed cycle, the parity argument (distance parity from a root is well defined and every arc flips it) would make G bipartite, but G contains triangles. So a kernel-perfect admissible orientation must contain odd directed cycles, none of which may be induced (an induced odd directed cycle has no kernel).

2 Structure of candidate orientations

Lemma 2. *In a kernel-perfect admissible orientation every column is a transitive triangle. Call its source S, middle M and sink T. Then both row arcs at S point into S, both row arcs at T point out of T, and M has one row arc in and one out.*

Proof. A column is an induced triangle. A directed triangle $x \to y \to z \to x$ has no kernel: its independent sets are singletons, and $\{x\}$ does not absorb y (the only arc out of y goes to z). So each column is transitive. S has both column arcs out, so its remaining out-degree is 0; T has both column arcs in, so both row arcs go out; M has one column arc out, so exactly one row arc out. □

Lemma 3 (Encoding)**.** *Let $e_r(c) = +1$ if the row-r arc between columns c and $c+1$ points right and -1 otherwise. Admissible orientations with all columns transitive correspond bijectively to pairs (ε, p) with $\varepsilon \in \{\pm 1\}$ and p a proper 3-colouring of the cycle C_n (so $p_c \in \mathbb{Z}_3$, $p_c \neq p_{c+1}$): at boundary c all three row arcs have sign ε except row p_c. For $\varepsilon = +1$, column c has $T = p_{c-1}$, $S = p_c$ and M the remaining row. There are $2(2^n - 2)$ such orientations.*

Proof. By Lemma 2, row r is the source S of column c iff $(e_r(c-1), e_r(c)) = (+,-)$, the sink T iff it is $(-,+)$, and M iff the two signs agree. Each column has exactly one of each, so passing from boundary $c-1$ to boundary c exactly one coordinate of (e_0, e_1, e_2) changes $+ \to -$ and one changes $- \to +$. The number of $+$ signs is therefore the same at every boundary, and it is 1 or 2 since a change needs both signs present. Let ε be the majority sign and p_c the unique row with the minority sign at boundary c. The row changing $+ \to -$ at column c is p_c and the row changing $- \to +$ is p_{c-1}; these differ, so p is proper. Conversely any (ε, p) defines column types with degrees $2, 2$ everywhere. The number of proper 3-colourings of C_n for odd n is $2^n - 2$. □

Reversing the column order maps $\varepsilon = -1$ to $\varepsilon = +1$ and preserves both kernel-perfectness and induced cycles, so from now on $\varepsilon = +1$.

Definition 4. Write $\delta_c = p_{c+1} - p_c \in \{1, 2\} \pmod 3$. The triple centred at c, (p_{c-1}, p_c, p_{c+1}), is *rainbow* if its entries are distinct and of type *aba* if $p_{c+1} = p_{c-1}$. It is rainbow iff $\delta_{c-1} = \delta_c$. We call p *Latin* if every triple is rainbow.

If p is Latin then δ is constant, $\sum_c \delta_c = n\delta \equiv 0 \pmod 3$, so $3 \mid n$. The number A of aba triples equals the number of sign changes of the cyclic sequence δ, hence is even; the number of rainbow triples $R = n - A$ is odd.

For $n = 3$ every proper colouring is Latin, the pattern of types is a Latin square of order 3 on $\{S, M, T\}$, and there are 12 such orientations.

3 $C_3 \square C_3$

Theorem 5. *$C_3 \square C_3$ has exactly 12 kernel-perfect admissible orientations, namely the Latin-square orientations.*

Proof. By Lemmas 2 and 3 only these 12 are candidates. In each, every column and every row is a transitive triangle. Colour column arcs red and row arcs blue. Each colour class is a disjoint union of transitive tournaments, so it is transitive, and this survives in induced subdigraphs. By the Sands–Sauer–Woodrow theorem every 2-arc-coloured digraph has a set S with no monochromatic path between two of its vertices such that every other vertex has a monochromatic path into S. With transitive colour classes, monochromatic paths are arcs, so S is a kernel. □

4 The obstruction: induced odd winding cycles

A *winding cycle* is given by rows b_c ($c \in \mathbb{Z}_n$) with $b_c \neq p_c$. In column c it uses the entry vertex (b_{c-1}, c) and, if $b_c \neq b_{c-1}$, also (b_c, c); it crosses from column c to $c+1$ along row b_c, which points right since $b_c \neq p_c$. A column where $b_{c-1} \neq b_c$ is a *step*.

Lemma 6. *Let $n \geq 5$. Suppose for every c: (W1) $b_c \neq p_c$; (W2) at a step, $(b_{c-1}, c) \to (b_c, c)$ is an arc, i.e. goes in the order $S \to M \to T$; (W3) at a step, $b_{c+1} \neq b_{c-1}$. Then the winding cycle is an induced directed cycle of length $n + \#steps$.*

Proof. It is a directed closed walk by (W1),(W2) and visits distinct vertices. Within a column it uses at most two vertices, joined by the arc it uses. Between columns c and $c+1$ the rows used are $\{b_{c-1}, b_c\}$ and $\{b_c, b_{c+1}\}$; the only shared row besides the crossing row b_c would be $b_{c-1} = b_{c+1}$ at a step, excluded by (W3). Columns at distance ≥ 2 are non-adjacent since $n \geq 5$. □

Rule A. Let $b_c = \overline{p_c p_{c+1}}$, the third row. In column c the roles are $T = p_{c-1}$, $S = p_c$, $M = \overline{p_{c-1} p_c} = b_{c-1}$, so the cycle always enters at M. If the triple at c is aba then $b_c = b_{c-1}$ (no step). If it is rainbow then $p_{c+1} = b_{c-1}$, so $b_c = \overline{p_c\, b_{c-1}} = p_{c-1} = T$: a step $M \to T$. (W3) holds because at a rainbow c, $b_{c-1} = p_{c+1} \neq b_{c+1}$. So Rule A gives an induced directed cycle of length $n + R$, which is *even*. A parity flip is needed.

Theorem 7. *Let $n \geq 5$ be odd and p not Latin. Then $D(\varepsilon, p)$ contains an induced odd directed cycle, so it is not kernel-perfect.*

Proof. Since p is not Latin, $A \geq 1$, and R is odd so $R \geq 1$. Going around the cycle there is therefore a rainbow triple immediately followed by an aba triple: an index c with $a = p_c$, $b = p_{c+1}$, $t = p_{c+2}$ distinct and $p_{c+3} = b$. Take Rule A and change one value: $b_c := b$ instead of t. Rule A gives $b_{c-1} = \overline{p_{c-1} a}$, $b_{c+1} = a$, $b_{c+2} = a$.

Column c has roles $T = p_{c-1}$, $S = a$, $M = b_{c-1}$, and $p_{c-1} \in \{b, t\}$. If $p_{c-1} = b$, Rule A stays at $M = t$ and the new cycle steps $M \to T = b$. If $p_{c-1} = t$, Rule A steps $M \to T = t$ and the new cycle stays at $M = b$. Either way (W2) holds and the number of steps changes by exactly one.

Column $c+1$ has roles $T = a$, $S = b$, $M = t$. Rule A steps $t \to a$; the new cycle steps $b \to a$, i.e. $S \to T$. Same count.

(W1): $b \neq a$. *(W3)*: at column $c+1$, $b_{c+2} = a \neq b$. At column c (if a step), $b_{c+1} = a \neq b_{c-1} = \overline{p_{c-1} a}$. At column $c-1$ (if a step), we need $b \neq b_{c-2} = \overline{p_{c-2} p_{c-1}}$: if $p_{c-1} = b$ this is automatic; if $p_{c-1} = t$, column $c-1$ is a step only when $p_{c-2} = b$, and then $b_{c-2} = a$. All other columns are unchanged.

So the new winding cycle is induced with $R \pm 1$ steps, an even number, and its length $n + R \pm 1$ is odd. An odd directed cycle has no kernel. □

Corollary 8. *For odd n with $3 \nmid n$, no admissible orientation of $C_3 \square C_n$ is kernel-perfect. For odd n with $3 \mid n$, the only possible ones are the 12 Latin orientations ($p_c = p_0 \pm c$, two choices of ε, three of p_0).*

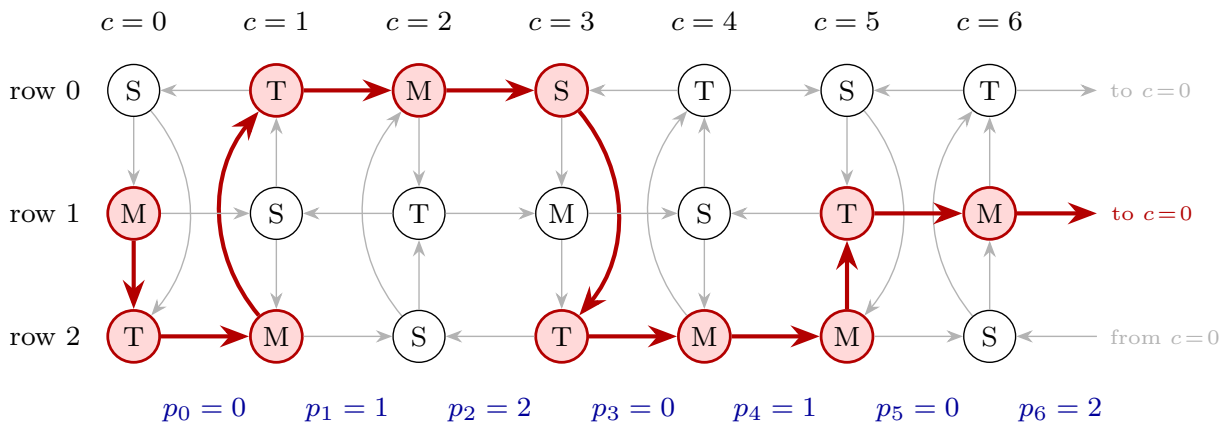


Figure 1: $n = 7$, $p = (0, 1, 2, 0, 1, 0, 2)$, column labels show S, M, T. The gadget is at $c = 2$; the highlighted induced directed cycle has length 11 (row arcs leaving column 6 wrap to column 0).

5 The Latin orientations are kernel-perfect (computer-assisted)

Theorem 9. *For every odd k, the 12 Latin orientations of $C_3 \square C_{3k}$ are kernel-perfect. Hence the kernel method proves $C_3 \square C_{3k}$ is 3-choosable.*

Method. Kernel existence in an induced subdigraph H is checked column by column with a 20-state automaton: the state after column c records the kernel vertex of column c (at most one, since columns are cliques) and which present vertices of column c still need absorbing through a row arc to column $c+1$. Every H is a choice of present rows in each column; one period of 3 columns gives 8^3 boolean 20×20 block matrices, and H on $C_3 \square C_{3k}$ has a kernel iff the product of its k blocks has a nonzero boolean trace. We computed, for each of the 12 orientations, the complete set of pairs (product, parity of k) closed under multiplication by blocks (1515 or 1604 pairs). No product with k odd has zero trace, which proves the claim for every odd k. □

Checks. The automaton agreed with direct kernel search on 37,800 induced subdigraphs. Independently, all 2^{27} induced subdigraphs of each Latin orientation of $C_3 \square C_9$ were checked by brute force. For even k the claim fails: on $C_3 \square C_6$ there is a 9-vertex kernel-less induced subdigraph, found by both methods.

Theorem 10 (Summary for $m = 3$)**.** *For odd $n \geq 3$, $C_3 \square C_n$ has a kernel-perfect orientation with $\Delta^+ \leq 2$ if and only if $3 \mid n$, and then exactly 12.*

6 $m \geq 5$

Proposition 11 (computer-verified)**.** *$C_5 \square C_5$ has no kernel-perfect admissible orientation: all 143,224 admissible orientations contain an induced odd directed cycle.*

Lifting the $C_3 \square C_3$ Latin orientations in both coordinates also fails: all 12 lifts to $C_9 \square C_9$ contain induced odd directed cycles.

Conjecture 12. *For odd $m, n \geq 5$, $C_m \square C_n$ has no kernel-perfect orientation with $\Delta^+ \leq 2$.*

7 Related verified facts

- **Edge deletion, exact.** For $C_5\Box C_5$ and $C_5\Box C_7$, every coefficient $[x_w \prod_{v\neq w} x_v^2]\, P_{G-e}$ with w off the cycle of e was computed over $\mathbb{F}_{1000003}$ (Lasoń / Karasev–Petrov formula with cube roots of unity) and is nonzero, hence nonzero over $\mathbb{Z}$. So, as for $m = 3$, any bad 3-list assignment of these graphs has $L(u) \cap L(v) \neq \emptyset$ on every edge.
- **Independent check of the $m = 3$ coefficient formulas** (Theorem 1 of "Deleting one edge..."): all values agree for $n = 3, 5, 7, 9$.
- **List-colouring search.** No bad 3-list assignment of $C_3\Box C_5$ exists with a palette of 4 colours (SAT search, exhaustive); larger palettes and $C_5\Box C_5$ still running.